

Project Title:	Signs for Arboriculture – Collaborative Project
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Project Overview:

What were the goals and objectives of the project?	The goal of this project was to make 30 signs representing the trees of Essex North Shore's Arboriculture Maplewood Arboretum. Each sign contains the name of the tree in English, and an additional subtext that contains the name in Latin. Underneath, there is a machined QR code that links to a google document containing more information for the tree. Additionally, it contains a logo and the text "Maplewood Arboretum".
What were the criteria for the success of the project?	Due to the previous shortcomings of the previous signs, they quickly faded and wore off. As a result, these new signs are fully machined, and therefore are not printed and do not suffer from the same failures. These elements were mounted on an aluminum base through bolts and "thru holes" to mount it very firmly. The aluminum base contains a base with set screws for mounting to the ground at a specific angle. This was attached through tig welding.
Was the project completed according to expectations? Explain.	This project was completed according to expectations, and Essex Tech Arboriculture was very thankful and content with the project. As this was a favor rather than a requirement, it meant a lot to them, and to the values for our shop. They will be successfully mounted in the ground for many years of Essex Tech's future history.
Additional Comments:	These signs were a collaboration between us, Metal Fabrication, and partially Carpentry for Arboriculture.

Project Highlights:

What were the major successes?	One of the project's major successes was the QR codes. The QR codes were machined with a milling machine, so they will not wear out over time. They also contain error correction, so they are resistant to dirt or debris. They work perfectly and scan from a wide range of distances.
What methods, tools, or skills worked well?	QR Codes are very complex from a topological and graphical complexity perspective. As the white and black spots form contrast, machining a tool path with an End Mill or other tool will be difficult due the small size needed to achieve such resolution with traditional machining operations such as a pocket or profile. As a result, software such as Fusion360 will be very suboptimal at automating the toolpath generation of 30 different QR codes. Fusion360 works with different size tools, and can't break free from the more common workflow of machine operations. As a result, I took my previous skills I had as a hobby software developer (C++), and wrote a program that takes all 30 QR codes, and uses a QR code generation library that generates the code. I used this program to then generate the G-code (toolpath) for each of the QR codes in an automated way. Creating the QR codes from a website, converting all 30 of them into vector-based files, and importing all of them individually into Fusion360 by importing the image into the sketch is very time consuming and disgusting to think about. My program optimizes this by parsing a text file containing all of the tree names and links, and generates the QR code at the same time it generates the G-code. Since engraving tools and spot drills can have varying visible diameter due to their nature of having a depth that can be changed, they make for a great option as a tool for engraving QR codes. Small end mills can be expensive, but the engraving tool does not need to be small due to its ability to only have a tiny radial portion of the tool engraved into the material. My program has a "for" loop, which loops through each of the spots, or modules, on a QR code on the X and Y axis, and does simple arithmetic to calculate

	the corresponding X and Y positional coordinates for the machine. It accumulates a string based on these positional values with corresponding G-code syntax prefixed to their operations. The depth can be calculated by dividing the optimal diameter by 2, because the X is proportional to the Y with a 45 degree tool (as $\tan(45 \text{ deg}) = 1$). The division by 2 occurs because the diameter is double the radius. This program was also optimized to save on cycle time by keeping the rapid distance to a minimum.
What did you find to be useful to complete the project?	I've found that my accumulation of knowledge regarding software and mathematics to be very useful and essential to this project. C++ and software is not anywhere near the agenda, curriculum, or expectations for the shop program. However, it was necessary to avoid taking less optimal shortcuts or wasting time on this project. I've also found that the collaboration between me and the other people working on this project in the shop to be very beneficial. As I worked on all of the front informational plates, the others worked on getting all of the aluminum to size and ready to be assembled. Additionally, the collaboration from the machines in Metal Fabrication and some of Carpentry was necessary for some of the tasks that this project required. Most obviously, the surplus of help and education I have received over the course of my shop years has trained me to this level, and my patience and dedication to participation in the shop exhibits my abilities.
Additional Comments:	Fusion360 was used for all of the other elements of the front-facing plate, such as the text elements and the logo. Fusion360 is very useful and time-saving, but not for the QR codes. The material for the front-facing plate was a plastic that was dual-layer, which allows for contrast between two depths.

Project Challenges:

What elements of the project went wrong?	One of the failure points of the project was when the QR code was being machined at first. As I didn't want to wreck parts, we used a test part for an example. One of the initial struggles was getting the Z alignment on the tool to be correct. The QR code was not readable at first, since the tool went too deep and caused each circle that outlines the modules to be too big and therefore would be blending in with others. I struggled to find the cause of such misalignment, however I realized that a slight imperfections in the sharp tip of the tool would therefore cause the tool to feel friction too deep into the touching off process, so therefore making the Z location of the tool offset to be too deep, and therefore result in a larger diameter. I fixed this issue by estimating an offset and checking it every once in a while before cutting the entire QR code to prevent damage. This ended up being a non-issue.
What specific processes, tasks, or actions need changing? Why?	One of the things that was minor, but still suboptimal, with the project was the alignment of the tool on the part. The Z depth looks like it is changing throughout the part. Nearing the very end of the project, I figured out it was because the tool was just chipped and needed to be replaced. This was minor since the project still came out okay, but it could have been sharper and higher quality overall. Additionally, the set screw holes were very hard to access due to the angle between them and the base of the part. I wasn't the cause of this issue, but it still should be addressed.
How can you improve these processes in the future?	The set screw issue could have been improved upon if they were paying attention. However, more importantly, assisting and collaborating with them in a more effective way could have prevented this issue and their mistake. My expertise could have been used to guide them, so they can improve and learn. Also, the imperfections in the graphic could have been fixed by ensuring the condition of the tool before cutting the part. I came up with many different ideas on why it could happen, but the answer could just be as simple as a worn tool. Sometimes it is not good to overcomplicate things.
What were the key problem areas for you? Why?	The key problem areas for me throughout this project were having a vision and organized process before having ideas for individual elements. In other words, organization. I handled organization just fine, however it felt that I could have felt more comfortable and more confident with my process if I had more practice and with more experience.
List your technical challenges.	There were not many technical challenges for this project, however there are still a few minor points to consider. My programming went flawlessly, however it required figuring out a new library (QRCodeGen) which was tedious and annoying to read the documentation. I also had a few issues with coordinates at first since I realized that the way the machine handles the Y coordinate and the standard coordinate system for images is backwards, so I had to remember to flip it. Also, Fusion360 was very difficult for some of the other operations. As always, they were fixed with more practice.
Additional Comments:	The Arbor project was a very good opportunity for the students in Advanced Manufacturing to learn new skills and work together to achieve a common goal.

Post Project Tasks/Future Considerations:

What are the continued needs of the project? If any?	I do not think there are any continued needs of the project, however we would like to see this project become a reality very soon. Perhaps they might need more tools in the future to perfect the project, however longevity will not be an issue for the foreseeable future.
What actions did not get completed?	All of the actions have been completed for this project. There was not necessarily a time limit, since this was replacing all other actions we were doing in the shop, and

Why?	there is no true and set curriculum. The actual things that are going into the ground however were not completed, but that was expected and their responsibility to find and use the set screws to mount. This was not completed due to the lack of resources.
Additional Comments.	I hope to see these projects in the real Arboretum and working as educational signs.

Project Planning Phase: List the major elements of your project plan

Element	Lessons Learned/Comments
Brainstorming	After we received the assignment from Arbor, we were brainstorming how we were to get this done. As suggested by many of us, it would be ideal to get engravable plastic for the graphics.
Fusion360	Fusion360 helped model (CAD) and helped prepare the manufacturing (CAM) for the logo, text, and other graphical contents on the front face except the QR codes
Calculations	Before actually manufacturing the part, calculations were made using a calculator to line up all of the elements of the front face graphics together in a way that looks graphically correct. This was done in Fusion as well, but required some manual arithmetic for the QR codes, which were not completed in Fusion.
Manufacturing Process	The manufacturing process was designed so that singular operations were to be done on all 30 parts. That way, the planning and manufacturing for each of the operations can be done at different times, due to convenience. For example, we needed to wait on DMC to vectorize the logo before we used it.

Project Execution Phase: List the major elements of executing your project plan

Element	Lessons Learned/Comments
Machining: Touching off tools	For the logo, QR codes, titling elements, and other machining operations, it required touching the tools off the top of the part to ensure the machine knows the tools' relative position to the part.
Machining: Edgefinding	To set a G54 on the machine for each of these elements, edgefinding was used to find the edges of a part, and offsets were applied depending on where the elements lay on the part relative to the origin in Fusion360.
Welding: Tig welding	An extra additional operation that was performed on the part was where the base was welded on the circular mount with set screws. This involved a new learning experience since our shop does not primarily work with welding.

Manual machining operations	Some of the additional features of the part such as the mounting screws, the exact length of the aluminum pieces, the angle on the base, etc were performed as expected on a manual machine.

Other Factors: List the major project elements not covered by planning or execution

Element	Lessons Learned/Comments
Logo	The logo was vectorized by DMC (Design and Media Communications) to save time on our machining operations, as it was collaborative and effective.
Plastic	The plastic used for the engraving was cut up into pieces by the saw in carpentry, as our equipment was not designed for the task.